Plausible but Not Brain-Driven: Quantifying Prior Dominance in Neural Image Reconstruction

Yuki Kikuchi

Preprint — 2026

Abstract

Brain-to-image reconstruction can produce strikingly realistic images, but it remains unclear whether these reconstructions are actually driven by brain signals. Strong generative priors can produce plausible images even when brain activity is randomized—a failure mode that statistics computed without the presented stimulus, such as visual plausibility or feature statistics, cannot detect.

We introduce the **Degree of Brain Control (BC)**, defined as the ratio of within-category feature variance under shuffled versus preserved brain–stimulus correspondence. $\mathrm{BC} = 1$ indicates prior-dominated reconstruction, whereas $\mathrm{BC} > 1$ indicates that brain signals impose structure on the output. With balanced trials, BC is an affine function of the one-way ANOVA F statistic (the Calinski–Harabasz index); unlike F, it does not grow with the number of trials, which makes it a calibrated effect size.

Applied to the GOD dataset, real fMRI yields $\mathrm{BC} = 1.259$ while shuffled signals yield $\mathrm{BC} = 1.001$, despite identical feature norms and comparable visual quality. This dissociation replicates across all five subjects (paired $t = 7.44$, $p = 0.0017$). Identification accuracy against the true stimulus also detects shuffling, but BC captures structural degradation of decoded representations that is not detectable by identification accuracy alone ($d = 0.436$, $p = 0.033$).

BC provides an interpretable scale, independent of the number of trials, for quantifying how much brain signals constrain reconstruction beyond model priors.

Contents

1 Introduction **3**

2 Related Work **4**
- 2.1 Brain-to-Image Reconstruction . . . 4
- 2.2 Evaluation Metrics for Brain Decoding . . . 4
- 2.3 Role of Generative Priors . . . 5

3 Method: Degree of Brain Control **5**
- 3.1 Problem Setup . . . 5
- 3.2 Definition . . . 6
- 3.3 Interpretation . . . 6
- 3.4 Relationship to the ANOVA F Statistic . . . 6
- 3.5 Relationship to Identification Accuracy . . . 7
- 3.6 Computation . . . 7

4 Experiments **7**
- 4.1 Setup . . . 7
- 4.2 Experiment 1: Prior-Dominated Baseline . . . 8
- 4.3 Experiment 2: BC vs. Identification Accuracy . . . 9
- 4.4 Experiment 3: Image-Space Brain Control (Phase 2 BC) . . . 10
- 4.5 Experiment 4: Comparison with Existing Metrics . . . 10

5 Discussion **11**
- 5.1 What BC Measures—and What It Does Not . . . 11
- 5.2 Implications for Diffusion-Based Reconstruction . . . 12
- 5.3 BC Beyond the Feature Decoder . . . 13
- 5.4 Limitations . . . 13

6 Conclusion **14**

1 Introduction

Brain-to-image reconstruction has progressed from recovering coarse spatial patterns [8, 9] to producing images that appear to reflect fine-grained visual content [10–12]. This progress has been driven in large part by increasingly powerful generative models—GANs, VAEs, and diffusion models—that serve as the image synthesis stage of reconstruction pipelines. The reconstructed images are evaluated primarily on visual quality: perceptual similarity, identification accuracy, or pixel-level fidelity. By these measures, recent methods appear highly successful.

Yet visual quality is not the same as brain control. A generative model with a strong prior can produce realistic, category-consistent images even when its inputs carry no brain-specific information. Consider a decoder that maps fMRI signals to intermediate features, followed by a generator that reconstructs an image from those features. If the generator's prior is sufficiently strong, it will produce a natural-looking image regardless of whether the decoded features reflect the actual stimulus—simply because the prior constrains the output to the manifold of natural images. Under these conditions, reconstructions may look convincing while being dominated by the model prior rather than by brain signals. Metrics that need no reference—visual plausibility or statistics of the decoded features—describe the output itself and cannot tell whether brain activity determined it. Reference-based metrics—structural similarity (SSIM), perceptual distance (LPIPS, DreamSim), and identification accuracy—compare the output with the presented stimulus. They can reveal that a reconstruction is unrelated to its stimulus, but they report similarity to the target rather than a calibrated measure of how strongly brain activity constrains the output.

We make this failure mode concrete: when the correspondence between fMRI trials and stimuli is randomly shuffled, the resulting reconstructions have identical feature norms and comparable visual quality to those produced by genuine brain signals—yet by construction contain no stimulus-specific information. Statistics that do not use the stimulus cannot detect this shuffling; identification accuracy can, but only by comparison with the ground-truth stimulus representation. This motivates a different question: not *how realistic* is the reconstruction, but *how much is the reconstruction controlled by the brain?*

To answer this question, we introduce the **Degree of Brain Control (BC)**, a metric defined as the ratio of within-category feature variance under shuffled versus preserved brain–stimulus correspondence. When brain signals carry no category-relevant information, shuffling does not change the variance structure of decoded features, and BC ≈ 1. When brain signals impose consistent structure within categories, preserved correspondence yields lower within-category variance than shuffled correspondence, and BC > 1. BC $= 1$ thus provides an empirically grounded baseline for prior-dominated reconstruction, and deviations above 1 are directly interpretable as evidence of brain influence. With the same number of trials per category, BC is an affine function of the one-way ANOVA F statistic computed on the decoded features, i.e. the Calinski–Harabasz index [1]. BC is therefore not a new statistic. Its value lies in reading this quantity as an effect size with a fixed null value of 1 that, unlike F, does not grow with the number of trials, and in using it to test reconstruction pipelines for prior dominance.

We make the following contributions:

1. **Identifying a fundamental failure mode.** We show that prior-dominated reconstruction, which yields plausible-looking output with no genuine brain influence, is undetectable by statistics that do not use the presented stimulus. This failure mode becomes more severe as generative priors grow stronger, making it increasingly relevant to the field.

2. **BC as a calibrated measure of brain control.** We define BC, show that it is an

affine function of the ANOVA F statistic (Calinski–Harabasz index) whose scale, unlike that of F, does not depend on the number of trials, and give its information-theoretic interpretation (per feature dimension, $I(c; \hat{x}_k) \leq \frac{1}{2} \log \mathrm{BC}_k$) and its relationship to decoder SNR. BC = 1 provides an interpretable floor for prior-dominated reconstruction.

3. **Empirical validation across five subjects.** Real fMRI yields BC = 1.259; shuffled signals yield BC = 1.001—despite identical feature norms and comparable visual quality. This dissociation replicates across all five subjects (paired $t = 7.44$, $p = 0.0017$), ruling out subject-specific effects.

4. **Comparison with existing metrics.** Feature statistics cannot separate real from shuffled input; identification accuracy can. BC captures structural degradation of decoded representations not detectable by identification accuracy alone. Matched-accuracy conditions differ significantly in BC ($d = 0.436$, $p = 0.033$), and BC degrades faster than accuracy under noise injection—demonstrating that BC and accuracy are complementary, not equivalent.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 formally defines BC. Section 4 presents our experiments. Section 5 discusses implications and limitations.

2 Related Work

2.1 Brain-to-Image Reconstruction

Early work on neural image reconstruction focused on recovering low-level visual features—such as oriented gratings and spatial frequency maps—directly from fMRI responses using linear models [8, 9]. A key advance came with the introduction of intermediate feature representations: rather than predicting pixels directly, Horikawa and Kamitani [6] decoded deep CNN features (AlexNet-type layers) from brain signals, introducing the Generic Object Decoding (GOD) dataset, and Shen et al. [12] reconstructed images from decoded features, optionally constraining them with a deep generator network trained on natural images. This approach produced recognizable object-level reconstructions for the first time at scale.

Subsequent work shifted toward more powerful generative backbones. Brain-Diffuser [10] and MindEye [11] leverage latent diffusion models conditioned on CLIP embeddings decoded from fMRI, achieving high perceptual quality and identification accuracy. MinD-Vis [2] similarly exploits masked brain modeling as a pretraining objective before diffusion-based synthesis. While these methods represent impressive engineering achievements, they share a common structural property: a powerful generative prior whose output quality is largely determined by the prior itself, independent of the quality of the brain-decoded input. Our work addresses a measurement gap that becomes increasingly important as these priors grow stronger.

2.2 Evaluation Metrics for Brain Decoding

Reconstruction quality in brain decoding is typically assessed along two axes: pixel-level fidelity and semantic accuracy.

Pixel-level metrics (MSE, SSIM, PSNR) compare reconstructed images to target images at the pixel level. These metrics are well-understood but poorly suited to high-level generative pipelines where the output lies on a low-dimensional natural-image manifold regardless of input quality.

Perceptual similarity metrics (LPIPS [15], DreamSim [3]) measure distance in deep feature spaces and better reflect human perceptual judgments. However, they still assess how similar the reconstruction *looks* to the target, not whether the brain signal determined that appearance.

Identification accuracy (top-k or leave-one-out ranking in feature space) tests whether the reconstruction is closer to its target image than to distractors. This has become a standard benchmark [11, 12]. Identification accuracy falls to chance when the output is unrelated to the stimulus, so it does detect a decoder that ignores brain activity. However, it requires the ground-truth stimulus representation, and its scale depends on the distractor set and saturates at high signal-to-noise ratio.

These metrics report how close a reconstruction is to its stimulus. None provides a calibrated measure of a more fundamental quantity: *to what degree are brain signals controlling the reconstruction, rather than the model prior?* BC is designed to answer this question.

Variance-ratio statistics. The ratio of between- to within-group variance underlies the one-way ANOVA F statistic and the Calinski–Harabasz index of cluster validity [1]. BC belongs to this family: with balanced trials it is an affine function of F (Section 3.4). Our contribution is to read this quantity as a calibrated effect size of brain control that does not depend on the number of trials, and to apply it to test reconstruction pipelines for prior dominance.

2.3 Role of Generative Priors

The tension between model priors and input fidelity is well-recognized in the generative modeling literature. In VAEs [7], the KL regularization term explicitly pulls latent codes toward a standard Gaussian prior, reducing the influence of individual inputs. In GANs [4] and diffusion models [5, 13], the prior is implicit but strong: sampling procedures are designed to produce outputs on the natural-image manifold. As model capacity increases, these priors become more constraining relative to the decoder input.

In the brain decoding context, this tension manifests as a trade-off between visual quality and brain control: stronger priors yield more realistic reconstructions but may reduce the degree to which individual brain states are reflected in the output. Several authors have noted informally that diffusion-based reconstructions may be "over-smoothed" toward category prototypes [10], but no quantitative metric has been proposed to measure this effect. BC formalizes this intuition as a measurable, interpretable quantity.

Concurrent work on hallucination detection in neural decoding [14] shares the motivation of identifying when generative models produce outputs not supported by the input signal, but focuses on linguistic decoding rather than image reconstruction and does not provide a variance-ratio formulation comparable to BC.

3 Method: Degree of Brain Control

3.1 Problem Setup

Let $\mathbf{b}_i \in \mathbb{R}^V$ denote the fMRI response on trial i, and let $c_i \in \{1, \ldots, C\}$ denote the category label of the presented stimulus. A reconstruction pipeline consists of two stages: a decoder $f : \mathbb{R}^V \to \mathbb{R}^d$ that maps brain signals to an intermediate feature vector, and a generator $g : \mathbb{R}^d \to \mathcal{I}$ that maps features to image space. In practice, f is a linear (Ridge regression) decoder trained on held-out data, and the decoded feature $\hat{\mathbf{x}}_i = f(\mathbf{b}_i) \in \mathbb{R}^d$ is used as input to g.

We evaluate brain control at the feature level, i.e., on the set of decoded features $\{\hat{\mathbf{x}}_i\}$ over N test trials. Let $S_c = \{i : c_i = c\}$ be the index set of trials belonging to category c,

with $|S_c| = n$ for all c.

3.2 Definition

We define two variance quantities over the decoded features.

Within-category variance under preserved correspondence (V_{pres}) measures trial-to-trial variability within the same category when brain signals are correctly paired with stimuli:

$$V_{\text{pres}} = \frac{1}{C} \sum_{c=1}^{C} \frac{1}{d} \sum_{k=1}^{d} \text{Var}_{i \in S_c}[\hat{x}_{ik}]. \tag{1}$$

Within-category variance under broken correspondence (V_{brok}) measures the same quantity after randomly permuting the trial indices, destroying the brain–stimulus pairing:

$$V_{\text{brok}} = \mathbb{E}_{\pi} \left[\frac{1}{C} \sum_{c=1}^{C} \frac{1}{d} \sum_{k=1}^{d} \text{Var}_{i \in S_c}[\hat{x}_{\pi(i),k}] \right], \tag{2}$$

where π is a uniformly random permutation of $\{1, \ldots, N\}$. The expectation is estimated by averaging over $N_{\text{shuf}} = 1000$ independent permutations.

The **Degree of Brain Control** is defined as:

$$\boxed{\text{BC} = \frac{V_{\text{brok}}}{V_{\text{pres}}}} \tag{3}$$

3.3 Interpretation

$\text{BC} = 1$ (prior-dominated): Shuffling trial indices does not change the variance structure of decoded features. The decoder output carries no category-specific information—reconstruction is determined entirely by the model prior, not by the brain signals.

$\text{BC} > 1$ (brain-controlled): Preserved correspondence yields lower within-category variance than broken correspondence. Trials within the same category produce more similar decoded features when correctly paired with their stimuli than when randomly paired. This indicates that brain signals impose consistent structure on the reconstruction.

The quantity $\text{BC} - 1$ serves as an effect-size measure of brain influence. Under a Gaussian approximation with shared within-category variance, BC bounds the mutual information between the category label and each feature dimension k:

$$I(c; \hat{x}_k) \leq \frac{1}{2} \log \text{BC}_k, \tag{4}$$

where BC_k is BC computed on dimension k alone. The bound holds because a Gaussian has maximal entropy for a given variance, and it is close to tight when BC is near 1, as in our data. $\text{BC} = 1$ corresponds to zero mutual information; $\text{BC} > 1$ corresponds to positive information transmission from stimulus category through brain signals to decoded features. For the full feature vector the corresponding bound involves determinants of covariance matrices rather than traces, so $\frac{1}{2} \log \text{BC}$ should not be read as the mutual information of the whole vector.

3.4 Relationship to the ANOVA F Statistic

Let SS_W, SS_B and $SS_T = SS_W + SS_B$ denote the within-category, between-category and total sums of squares of the decoded features, summed over feature dimensions. The sample variance of a subset drawn without replacement is unbiased for the finite-population variance.

When $|S_c| = n$ for all c, the permutation expectation in Eq. (3) therefore has a closed form, $V_{\text{brok}} = SS_T/(d(N-1))$ and $V_{\text{pres}} = SS_W/(d(N-C))$, so that

$$\text{BC} = \frac{N-C}{N-1} \cdot \frac{SS_T}{SS_W} = 1 + \frac{C-1}{N-1}(F-1), \qquad F = \frac{SS_B/(C-1)}{SS_W/(N-C)}. \tag{5}$$

F is the one-way ANOVA F statistic computed on traces, which equals the Calinski–Harabasz index [1] of the decoded features with categories as clusters. BC therefore carries the same information as F; permutations are needed only for a p-value. The two differ in scale. For a fixed ratio of between- to within-category variance, F grows roughly in proportion to n, whereas $\text{BC} \to 1 + \operatorname{tr}\Sigma_B / \operatorname{tr}\Sigma_W$ as $n \to \infty$ and does not depend on n. BC relates to F as an effect size relates to its test statistic; we verify this empirically in Section 4.5.

3.5 Relationship to Identification Accuracy

Identification accuracy (e.g., leave-one-out cosine similarity ranking) is a commonly used metric in brain decoding. Both BC and accuracy increase with the signal-to-noise ratio of the decoder. However, identification accuracy is a nonlinear (sigmoidal) function of SNR and saturates at high SNR, whereas BC scales approximately linearly:

$$\text{BC} \approx 1 + \text{SNR}_{\text{eff}}, \tag{6}$$

where $\text{SNR}_{\text{eff}} = \operatorname{tr}\Sigma_B / \operatorname{tr}\Sigma_W$ is the ratio of between- to within-category variance in feature space (Section 3.4). This means BC continues to grow in regimes where accuracy has already saturated, enabling detection of structural differences that accuracy cannot resolve. We demonstrate this empirically in Section 4.3.

3.6 Computation

Given N test trials with decoded features $\{\hat{\mathbf{x}}_i\}$ and category labels $\{c_i\}$:

1. Compute V_{pres} directly from the decoded features.
2. Estimate V_{brok} by averaging over $N_{\text{shuf}} = 1000$ random permutations.
3. Compute $\text{BC} = V_{\text{brok}}/V_{\text{pres}}$.

Experiments 1 and 4 use the ratio of pooled variances in Eq. (3). Experiment 2 computes BC per category, BC_c, and averages across categories; the per-category values are used for the effect-size analysis (Cohen's d), and their standard error across categories serves as the uncertainty estimate.

4 Experiments

4.1 Setup

Dataset. We use the Generic Object Decoding (GOD) dataset [6], which contains fMRI responses from five subjects viewing 1,200 training images and 50 test images (one image per category, repeated 35 times each). Test trials thus consist of $50 \times 35 = 1{,}750$ trials per subject. All analyses use the Visual Cortex (VC) ROI unless otherwise stated.

Decoder. We train a Ridge regression decoder ($\alpha = 100$) with standard scaling to predict DNN features from brain signals. Experiment 1 and its multi-subject replication use the `cnn8` features distributed with the GOD dataset (final fully connected layer of an AlexNet-type network, $d = 1{,}000$), which are available for all 1,200 training images. Experiment 2 and

the reconstructed images use AlexNet relu7 features (fc7 after ReLU, torchvision ImageNet weights, $d = 4{,}096$), which serve as input to the relu7generator; these were extracted for the 733 of 1,200 training images available to us, so the corresponding decoders are trained on those 733 trials. Decoders are trained on training-set trials and evaluated on held-out test trials, following the original GOD experimental protocol [6].

BC computation. For each experiment, BC is computed on decoded features in their original units using $N_{\text{shuf}} = 1{,}000$ across-trial permutations (Eq. (3)); see Section 3 for the estimator used in each experiment.

4.2 Experiment 1: Prior-Dominated Baseline

Setup. We compare three conditions that differ only in the brain signals fed to the decoder:

- **Real**: genuine fMRI test signals (correct brain–stimulus pairing).
- **Shuffled**: test trial indices randomly permuted before decoding, destroying brain–stimulus correspondence while preserving the marginal distribution of brain signals.
- **Random**: standard Gaussian noise substituted for brain signals, representing complete absence of brain information.

The decoder and generator are identical across all three conditions.

Results.

Table 1: BC, feature norm and pairwise identification accuracy across three input conditions (Subject 1, VC ROI, `cnn8` features).

Condition	BC	Feature norm (L2)	Identification accuracy
Real	**1.259**	70.9	**0.726**
Shuffled	1.001	70.9	0.500
Random	1.000	118.4	0.490

The key finding is the dissociation between BC and feature norm (Table 1). Real and Shuffled conditions produce decoded features with identical L2 norms (70.9)—the shuffled set contains the same decoded vectors in a different order—meaning that any metric based on feature magnitude—and by extension, visual quality—cannot distinguish them. Yet BC separates them sharply: Real yields BC = 1.259, while Shuffled yields BC = 1.001, consistent with pure prior domination (BC = 1 by construction when correspondence is destroyed). Identification accuracy against the true stimulus features also separates them (0.726 vs. 0.500, chance 0.5); we compare BC with such metrics in Section 4.5.

Figure 1 shows reconstructed images for three representative categories under all three conditions. The Real and Shuffled rows are visually indistinguishable and all three look equally plausible, confirming that feature norm (and visual quality) is insufficient to distinguish brain-controlled from prior-dominated reconstruction. Figure 2 presents the BC and feature norm comparison as bar charts, making the dissociation explicit.

Multi-subject replication. To assess generalizability, we repeat Experiment 1 for all five subjects in the GOD dataset (Table 2). Real BC exceeds Shuffled BC in every subject without exception. A paired t-test across subjects yields $t = 7.44$, $p = 0.0017$, confirming that the Real–Shuffled dissociation is not specific to a single subject.



Figure 1: Reconstructed images under three input conditions (Real, Shuffled, Random) for three representative categories (goldfish, butterfly, bicycle), showing four repetitions each. The Real and Shuffled rows are visually indistinguishable despite the large difference in BC, demonstrating that visual quality cannot distinguish brain-controlled from prior-dominated reconstruction. Images are generated by the relu7generator from relu7 decoded features; BC values refer to the `cnn8` decoder (Table 1).

Table 2: BC across five subjects (VC ROI, `cnn8` features). Shuffled BC ≈ 1.000 in all subjects; feature norms of Real and Shuffled are identical within each subject.

Subject	Real BC	Shuffled BC	Feature norm (Real = Shuffled)
Subject 1	1.259	1.001	70.9
Subject 2	1.135	0.999	86.4
Subject 3	1.250	0.999	83.5
Subject 4	1.208	1.001	92.9
Subject 5	1.138	1.002	84.0
Paired t-test (Real vs. Shuffled): $t = 7.44$, $p = 0.0017$			

4.3 Experiment 2: BC vs. Identification Accuracy

Motivation. A natural concern is that BC may be redundant with identification accuracy. We investigate the relationship between the two metrics and examine whether BC provides information beyond what accuracy already captures.

ROI comparison. We compute both BC and leave-one-out cosine similarity accuracy across ten visual ROIs (V1, V2, V3, V4, LOC, FFA, PPA, LVC, HVC, VC) for Subject 1, using relu7 features. Across ROIs, BC and accuracy are highly correlated ($r = 0.976$), which is expected: both reflect the quality of the decoded representation and both are computed from the same Ridge decoder. This high correlation provides a basic validity check—BC is not measuring noise.

However, correlation does not imply equivalence. BC and accuracy measure different properties: accuracy captures the ability to rank the correct image above distractors, while BC measures the absolute variance compression imposed by brain signals within categories.

Same accuracy, different BC. To demonstrate the non-redundancy of BC directly, we inject Gaussian noise ($\sigma = 0.75$) into standardized V1 signals and identify the noise level at which V1+noise matches the identification accuracy of the HVC ROI. Despite being matched in accuracy, V1+noise and HVC differ significantly in BC (Cohen's $d = 0.436$, 95% CI $= [0.002, 0.041]$, $p = 0.033$, Welch's two-sample t-test over per-category BC values). Two conditions can have equivalent identification accuracy while differing in the structural quality of their decoded representations—a difference that BC captures and accuracy cannot.

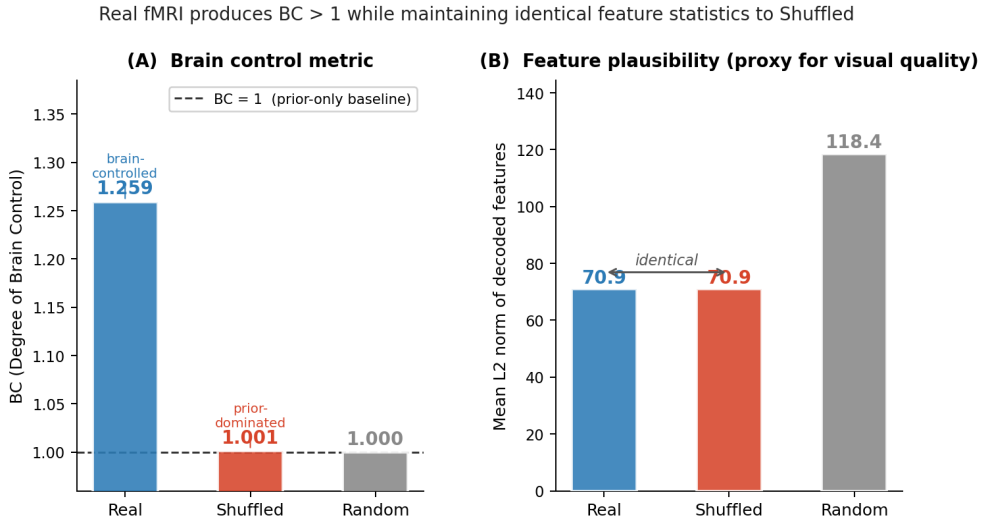


Figure 2: (A) BC and (B) feature L2 norm across three conditions. Real and Shuffled conditions have identical feature norms (right panel), yet BC distinguishes them sharply (left panel). The dashed line indicates BC = 1 (prior-only baseline).

Figure 3 shows the full noise sensitivity curves. BC degrades faster and more continuously than accuracy under noise injection, confirming the theoretical prediction that BC tracks SNR linearly while accuracy tracks it sigmoidally. Figure 4 shows the BC–accuracy scatter across ROIs, with the matched-accuracy comparison highlighted.

4.4 Experiment 3: Image-Space Brain Control (Phase 2 BC)

Setup. Experiments 1–2 measure BC in the decoded feature space (Phase 1 BC; `cnn8` in Experiment 1, relu7 in Experiment 2). To assess whether brain control extends to the final image space, we define Phase 2 BC using DreamSim embeddings [3] ($d = 1{,}792$) computed directly from generated images.

Results. Phase 2 BC across 50 categories has mean 1.036 ± 0.105 (std). The per-category distribution is wide: goldfish reaches BC = 1.299 while shovel reaches BC = 0.840 (below 1, indicating that the generator introduces category-inconsistent variation not present in feature space). The correlation between Phase 1 BC (`cnn8` decoder) and Phase 2 BC across categories is $r = 0.257$ ($p = 0.072$), a weak positive relationship that does not reach statistical significance.

The weak Phase 1–Phase 2 correlation suggests that feature-space brain control does not fully transfer to image-space brain control. This motivates measuring BC at both stages of the pipeline when evaluating reconstruction methods.

4.5 Experiment 4: Comparison with Existing Metrics

Setup. For the Real, Shuffled and Random conditions of all five subjects (`cnn8` features), we compute four quantities on the decoded features: (i) the mean L2 norm, which needs no reference; (ii) pairwise identification accuracy, which correlates each decoded feature vector with the true `cnn8` features of the 50 test images and reports the fraction of distractors ranked below the true image (chance 0.5); (iii) the Calinski–Harabasz index; and (iv) BC. To test the dependence on sample size, we draw $n = 5, 10, \ldots, 35$ trials per category (20 draws per n) from the Real condition of Subject 1.

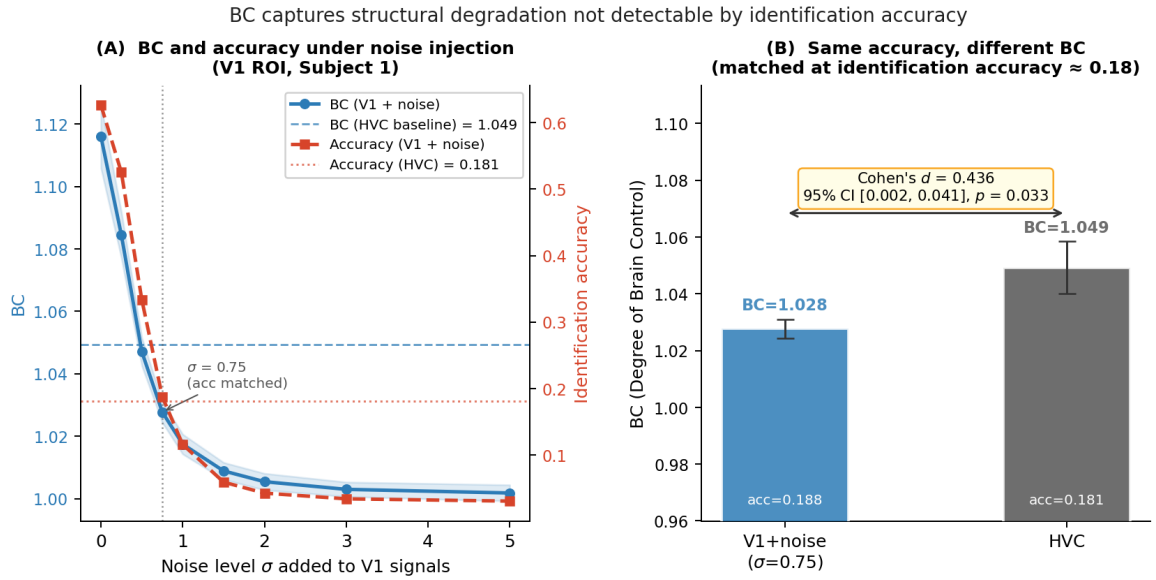


Figure 3: (A) BC (blue) and identification accuracy (red) as a function of noise level σ added to V1 signals, together with HVC baselines (dashed). BC degrades faster than accuracy. (B) At $\sigma = 0.75$, accuracy is matched between V1+noise and HVC, but BC differs significantly ($d = 0.436$, $p = 0.033$).

Table 3: Existing metrics and BC across input conditions (mean over five subjects, range in brackets; VC ROI, `cnn8` features).

Metric	Needs	Real	Shuffled	Random
Feature norm (L2)	nothing	83.5 [70.9, 92.9]	83.5 [70.9, 92.9]	94.0 [80.6, 118.4]
Identification accuracy	true stimulus	0.727 [0.697, 0.764]	0.499 [0.489, 0.505]	0.497 [0.489, 0.504]
Calinski–Harabasz	trial labels	8.06 [5.80, 10.25]	1.01 [0.95, 1.05]	1.02 [0.95, 1.09]
BC	trial labels	1.198 [1.135, 1.259]	1.000 [0.999, 1.002]	1.000 [0.999, 1.003]

Results. The feature norm cannot separate Real from Shuffled input in any subject (Table 3). Identification accuracy separates them in every subject (Real 0.697–0.764; Shuffled 0.489–0.505). The Calinski–Harabasz index and BC separate them as well and, as predicted by Eq. (5), carry the same information: across all 15 subject × condition cells, the 1,000-permutation estimate of BC matches the closed form to within 0.0002. They differ in scale (Fig. 5D). As n decreases from 35 to 5, the Calinski–Harabasz index falls from 10.25 to 2.31, whereas BC stays between 1.257 and 1.261. BC thus behaves as an effect size that can be compared across experiments with different numbers of repetitions, while F and the Calinski–Harabasz index behave as test statistics.

5 Discussion

5.1 What BC Measures—and What It Does Not

BC is a variance ratio, not a proportion of variance explained. A value of BC = 1.259 does not mean "25.9% of the reconstruction is brain-controlled." It means that within-category feature variance is 25.9% higher when brain–stimulus correspondence is destroyed than when it is preserved—that is, genuine brain signals make same-category reconstructions more similar to each other than random pairing would. BC − 1 is best interpreted as an effect-size measure of

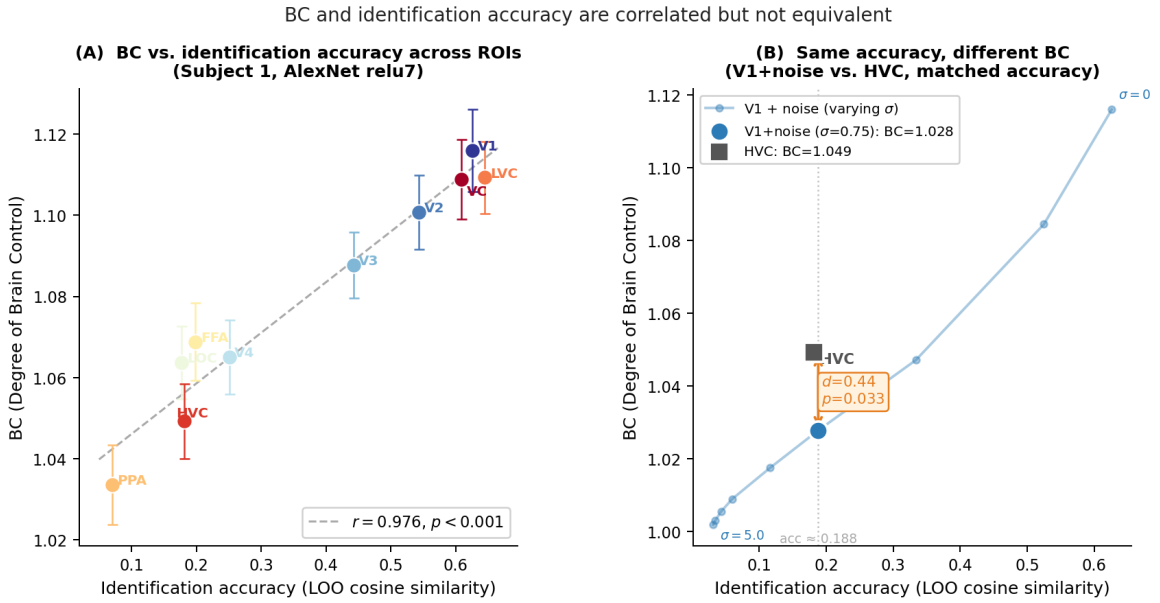


Figure 4: (A) BC vs. identification accuracy across ten visual ROIs ($r = 0.976$). (B) The matched-accuracy comparison: V1+noise and HVC have the same accuracy but different BC, demonstrating non-redundancy.

brain influence in feature space, directly analogous to a signal-to-noise ratio in the decoded representation.

Crucially, BC = 1 is not simply a null result; it is an empirically grounded floor. When brain signals carry no category-relevant information—as in the Shuffled condition—BC = 1 by construction, regardless of how realistic the reconstructions appear. Any reconstruction pipeline that yields BC $\approx$ 1 on genuine brain data should be considered prior-dominated, regardless of its visual quality.

BC is closely related to classical variance-ratio statistics. With balanced trials it is an affine function of the ANOVA F statistic and the Calinski–Harabasz index (Section 3.4), so it contains no information that F lacks. What distinguishes BC is its scale: BC = 1 is the null value regardless of the design, and BC $-$ 1 approximates the ratio of between- to within-category variance independently of the number of trials (Section 4.5). This makes BC values comparable across experiments with different numbers of repetitions, which F values are not.

5.2 Implications for Diffusion-Based Reconstruction

The most visually impressive brain reconstruction methods today rely on latent diffusion models [10, 11]. These models have extremely strong priors—trained on billions of images and optimized to produce outputs precisely on the natural-image manifold. This is precisely the regime where BC is most informative: as the prior strengthens, the generator becomes increasingly capable of producing high-quality, category-consistent images from near-arbitrary inputs, and BC should approach 1 even for genuine brain signals.

To empirically ground this prediction, we simulate increasing prior dominance via feature mixing. We linearly interpolate decoded features between the Real condition ($\alpha = 0$) and the Shuffled condition ($\alpha = 1$, prior-dominated by construction):

$$\hat{\mathbf{x}}_\alpha = (1 - \alpha)\, \hat{\mathbf{x}}_{\text{real}} + \alpha\, \hat{\mathbf{x}}_{\text{shuffled}}. \tag{7}$$

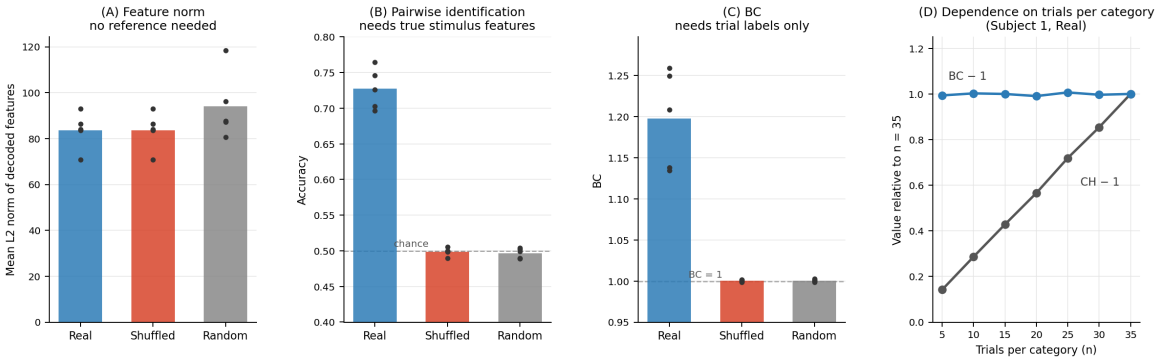


Figure 5: (A–C) Feature norm, pairwise identification accuracy and BC for Real, Shuffled and Random input (bars: mean over five subjects; dots: individual subjects). Dashed lines mark chance (B) and BC = 1 (C). (D) BC−1 and CH−1 as the number of trials per category varies, relative to $n = 35$ (Subject 1, Real).

As α increases from 0 to 1, BC decreases monotonically from 1.259 to 1.001 (Fig. 6A), with a near-linear trajectory (Fig. 6B). This demonstrates that BC is a continuous, sensitive measure of prior dominance: any generative process that partially replaces brain-specific information with prior-driven information will produce a BC value proportional to the degree of replacement.

This directly supports the prediction that diffusion-based reconstruction methods—whose stronger generative priors impose more structure on the output independently of the input—will yield BC values closer to 1 than the relu7generator used here, despite achieving higher perceptual quality and identification accuracy. Applying BC to publicly available diffusion-based reconstructions remains an important direction for future work.

5.3 BC Beyond the Feature Decoder

Our results show only a weak, non-significant correlation between Phase 1 BC and Phase 2 BC ($r = 0.257$, $p = 0.072$). This suggests that brain control in feature space does not fully transfer to image space—the generator introduces additional variability that can either amplify or dilute the brain signal depending on the category. Phase 2 BC, measured directly on generated images, captures complementary information about the full pipeline and should be reported alongside Phase 1 BC when the generator is not a simple deterministic inversion.

5.4 Limitations

Linear decoder. BC as computed here depends on the Ridge regression decoder. A more expressive decoder could extract more category-relevant information from brain signals, increasing BC. Our results represent a lower bound on the brain control achievable with the GOD dataset and AlexNet features.

Feature space selection. BC is measured in AlexNet-type feature spaces (`cnn8` and relu7). Different feature spaces may yield different BC values; in particular, higher-level semantic features (e.g., CLIP embeddings) may show higher BC because they are more directly predictable from high-level visual cortex activity. The choice of feature space should be reported when comparing BC values across studies.

Absolute scale. While BC provides an absolute scale relative to the prior-dominated baseline (BC = 1), the numerical value of BC − 1 is not directly comparable across datasets, decoders, or feature spaces. BC is most informative as a within-study comparison.

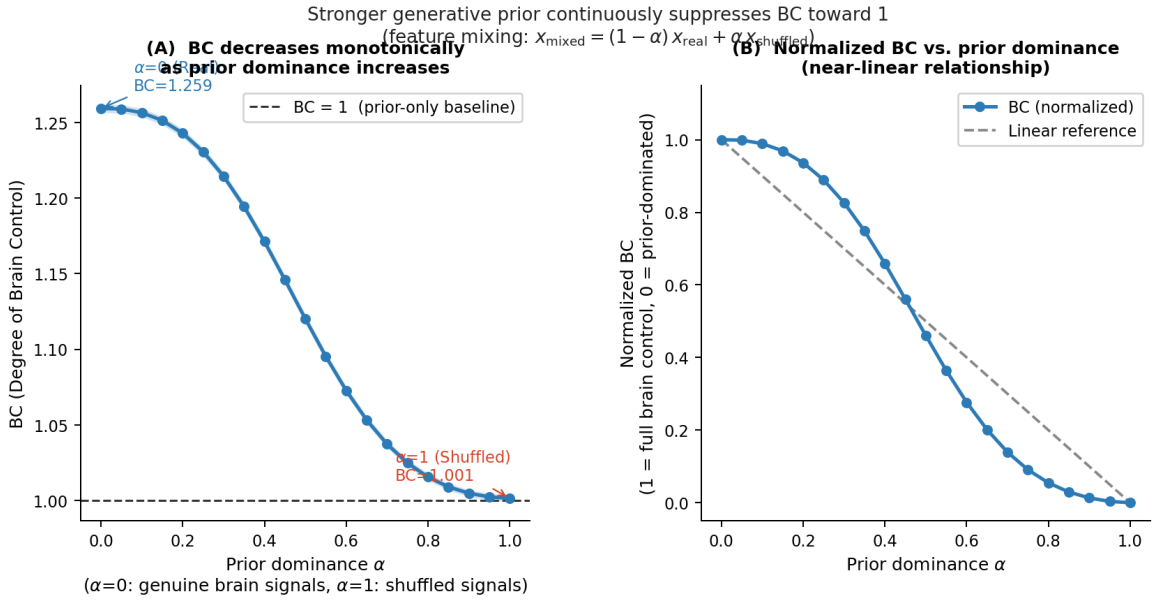


Figure 6: (A) BC as a function of prior dominance α, where $\hat{\mathbf{x}}_\alpha = (1-\alpha)\hat{\mathbf{x}}_{\text{real}} + \alpha\hat{\mathbf{x}}_{\text{shuffled}}$. BC decreases monotonically from 1.259 ($\alpha = 0$, full brain control) to 1.001 ($\alpha = 1$, prior-dominated). (B) Normalized BC shows a near-linear relationship with prior dominance, confirming that BC is a continuous, sensitive measure of the brain–prior trade-off.

6 Conclusion

Brain-to-image reconstruction has made remarkable progress, but visual quality alone cannot tell us whether the brain is actually driving the output. We introduced the Degree of Brain Control (BC), a metric that directly addresses this question by comparing within-category feature variance under preserved versus shuffled brain–stimulus correspondence. BC = 1 provides an empirically grounded floor corresponding to pure prior-dominated generation; BC $>$ 1 indicates that brain signals impose measurable structure on the reconstruction.

Applied to the GOD dataset, genuine fMRI signals yield BC = 1.259, while shuffled signals yield BC = 1.001—despite producing decoded features with identical norms and visually indistinguishable reconstructions. This dissociation, replicated across all five subjects ($t = 7.44$, $p = 0.0017$), shows that statistics computed without the presented stimulus cannot detect prior-dominated reconstruction. Identification accuracy against the true stimulus does detect it, but BC reveals structural degradation not detectable by identification accuracy alone ($d = 0.436$, $p = 0.033$), demonstrating that the two metrics are complementary rather than redundant. BC is an affine function of the ANOVA F statistic; unlike F, its scale does not depend on the number of trials, making it a calibrated effect size of brain control.

As generative priors in brain reconstruction grow stronger, the risk of conflating visual quality with brain control will only increase. We argue that BC should be reported alongside perceptual quality metrics in all brain-to-image reconstruction studies, and that maximizing BC—not just visual fidelity—should be an explicit goal of future decoder development.

References

[1] Tadeusz Caliński and Jerzy Harabasz. A dendrite method for cluster analysis. *Communications in Statistics*, 3(1):1–27, 1974. doi: 10.1080/03610927408827101.

[2] Zijiao Chen, Jiaxin Qing, Tiange Xiang, Wan Lin Yue, and Juan Helen Zhou. MinD-Vis: Unleashing cognitive synergy in visual decoding from the human brain. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19axis–19axis, 2023.

[3] Stephanie Fu, Netanel Tamir, Shobhita Sundaram, Lucy Chai, Richard Zhang, Tali Dekel, and Phillip Isola. DreamSim: Learning new dimensions of human visual similarity using synthetic data. In *Advances in Neural Information Processing Systems*, 2023.

[4] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems*, volume 27, 2014.

[5] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.

[6] Tomoyasu Horikawa and Yukiyasu Kamitani. Generic decoding of seen and imagined objects using hierarchical visual features. *Nature Communications*, 8:15037, 2017. doi: 10.1038/ncomms15037.

[7] Diederik P Kingma and Max Welling. Auto-encoding variational Bayes. *arXiv preprint arXiv:1312.6114*, 2014.

[8] Yoichi Miyawaki, Hajime Uchida, Okito Yamashita, Masa-aki Sato, Yusuke Morito, Hiroki C Tanabe, Norihiro Sadato, and Yukiyasu Kamitani. Visual image reconstruction from human brain activity using a combination of multiscale local image decoders. *Neuron*, 60(5):915–929, 2008.

[9] Shinji Nishimoto, An T Vu, Thomas Naselaris, Yuval Benjamini, Bin Yu, and Jack L Gallant. Reconstructing visual experiences from brain activity evoked by natural movies. *Current Biology*, 21(19):1641–1646, 2011.

[10] Furkan Ozcelik and Rufin VanRullen. Brain-diffuser: Natural scene reconstruction from fMRI signals using generative latent diffusion. In *arXiv preprint arXiv:2303.05334*, 2023.

[11] Paul S Scotti, Atmadeep Banerjee, Jimmie Goode, Jain Shailee, Andrew Gaibert, Tanvir Abraham, Idan Peer, Liuyi Luo, Hooroo Bheemreddy, Evelina Ivanova, et al. MindEye: Reconstructing the mind's eye: fMRI-to-image with contrastive learning and diffusion priors. In *Advances in Neural Information Processing Systems*, 2023.

[12] Guohua Shen, Tomoyasu Horikawa, Kei Majima, and Yukiyasu Kamitani. Deep image reconstruction from human brain activity. *PLOS Computational Biology*, 15(1):e1006633, 2019.

[13] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.

[14] Jerry Tang, Amanda LeBel, Shailee Jain, and Alexander G Huth. Semantic reconstruction of continuous language from non-invasive brain recordings. *Nature Neuroscience*, 26: 858–866, 2023.

[15] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 586–595, 2018.